

# Linear systems of three unknowns



1

a No

b Yes

c Yes

d No

e No

f No

2

a  $x = 1, y = 3, z = -2$

b  $x = 1, y = 5, z = 8$

c  $x = 4, y = -5, z = -2$

d  $x = -3, y = 5, z = -2$

e  $x = -3, y = 1, z = 2$

f  $x = 2, y = -1, z = -3$

g  $x = 4, y = -1, z = 3$

h  $x = 8, y = 10, z = 9$

i  $x = 4, y = 8, z = 6$

j  $x = 1, y = 2, z = -3$

3

a There are an infinite number of solutions.

b There are no solutions.

4

a  $y = t - 1$

b  $x = 2t - 11$

c  $x = 2t - 11, y = t - 1, z = t$

5

a  $y = -x - 2t + 9$

b  $x = -11t + 34$

c  $y = 9t - 25$

d  $x = -11t + 34, y = 9t - 25, z = t$

6

a  $x = 3y - 3t + 6$

b  $y = -t - 1$

c  $x = -6t + 3$

d  $x = -6t + 3, y = -t - 1, z = t$

7

a No solution

b Inconsistent

8

a  $x = 2, y = 5, z = 7$

b Independent.

9

a  $x = 0.2, y = 0.5, z = 0.1$

b Consistent

10 Example answer:

$$x + y + z = -7$$

$$x + y - z = -5$$

$$x - y + z = -3$$



11

a  $x = \frac{1}{6}, y = \frac{1}{3}, z = \frac{1}{4}$

b  $x = \frac{1}{3}, y = \frac{1}{6}, z = \frac{1}{2}$

12

a i  $x = 7, y = 4, z = 3$

ii Consistent

iii Independent

b i  $x = 4, y = 1, z = 6$

ii Consistent

iii Independent

c i  $x = 8, y = 8, z = 8$

ii Consistent

iii Independent

d i  $x = 19, y = 4, z = 16$

ii Consistent

iii Independent

e i  $x = 2, y = 3, z = 8$

ii Consistent

iii Independent

f i  $x = 13, y = 9, z = 1$

ii Consistent

iii Independent

g i  $x = 9, y = 7, z = 7$

ii Consistent

iii Independent

h i  $x = 3, y = 1, z = 9$

ii Consistent

iii Independent

i i  $x = 0.3, y = 0.5, z = 0.5$

ii Consistent

iii Independent

## Applications

13

a  $4A + 3B + 12C = 48$

b  $4A + 3B + 6C = 36$

c  $2A + B + C = 12$

d  $A = 3, B = 4, C = 2$

14



15

a  $a + b + c = 15$

$$4a + 2b + c = 5$$

$$16a + 4b + c = -3$$

c  $y = 2x^2 - 16x + 29$

b  $a = 2, b = -16, c = 29$

16

a  $2a - 4b + c = -20$

$$5a + b + c = -26$$

$$3a - 3b + c = -18$$

c  $x^2 + y^2 + 8x - 6y - 60 = 0$

b  $a = 8, b = -6, c = -60$